Welcome to the course

Git Basics



Course Overview

Welcome to the course on

Git Basics

This course provides overall understanding of the Git Basics and Branching Strategies. The knowledge gained from the course will help you understand how GIT is used in source code management.

At the end of this course, you should be able to:

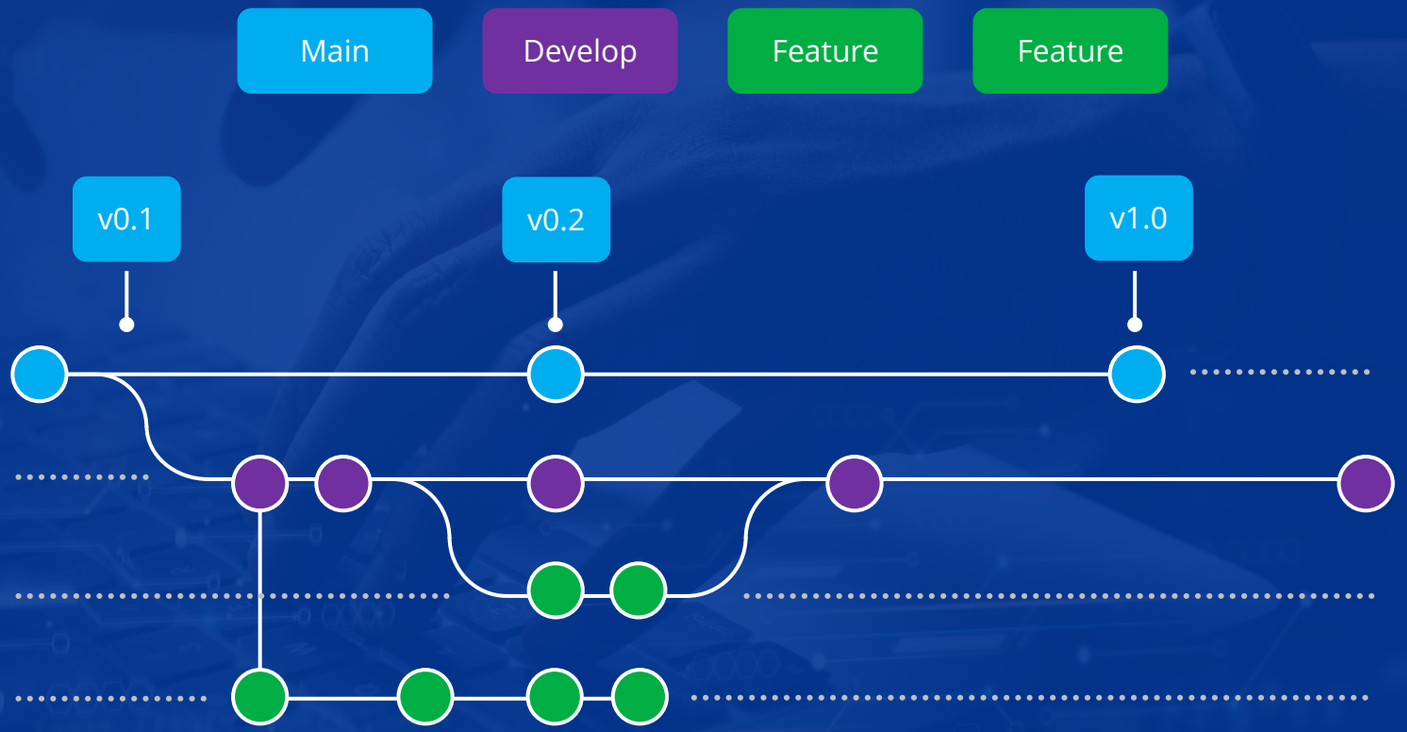
- Define GIT.
- Describe the basics of GIT that includes:
  - Version Control System (VCS)
  - Branching Strategies



What is a GIT?

- Development teams need a space to make changes, collaborate, and test new ideas without impacting the central codebase.
- They also need a way to move complete and properly tested features into the codebase.

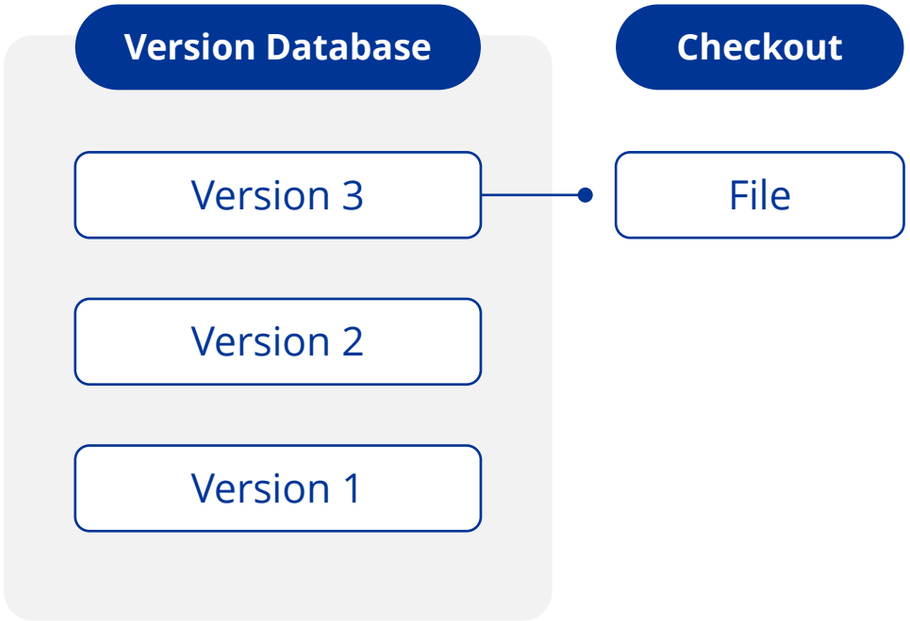
GIT is a collaborative workspace where developers can track the version of the source code.



Version Control System (VCS)

Version control system (VCS):

- Is a system that enables tracking and controlling the changes to files/documents.
- Helps developers to work on a shared codebase, thereby no burden of using shared files and folders.

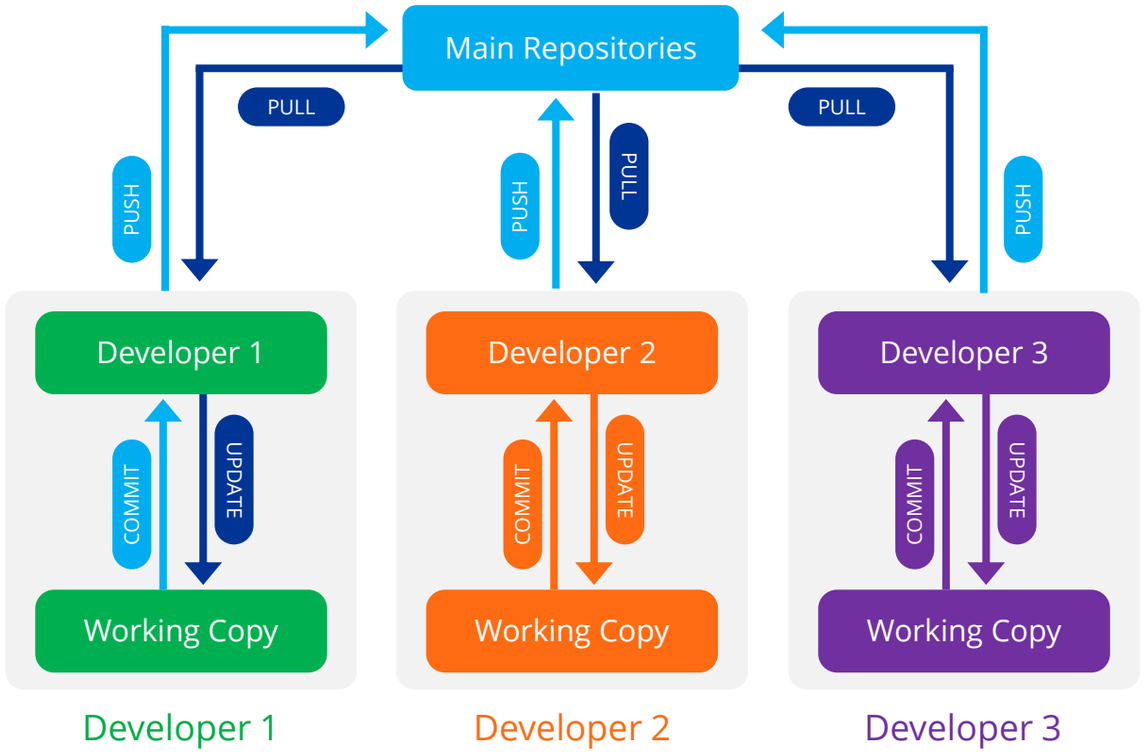


Version Control System (VCS)

There are two primary aspects of version control systems:

Repositories

- Repositories are the core of the VCS housing the edited code, while developers review and make further updates.
- This storage system allows for clarity and collaboration across development teams through the process of committing and updating.



Working Copy

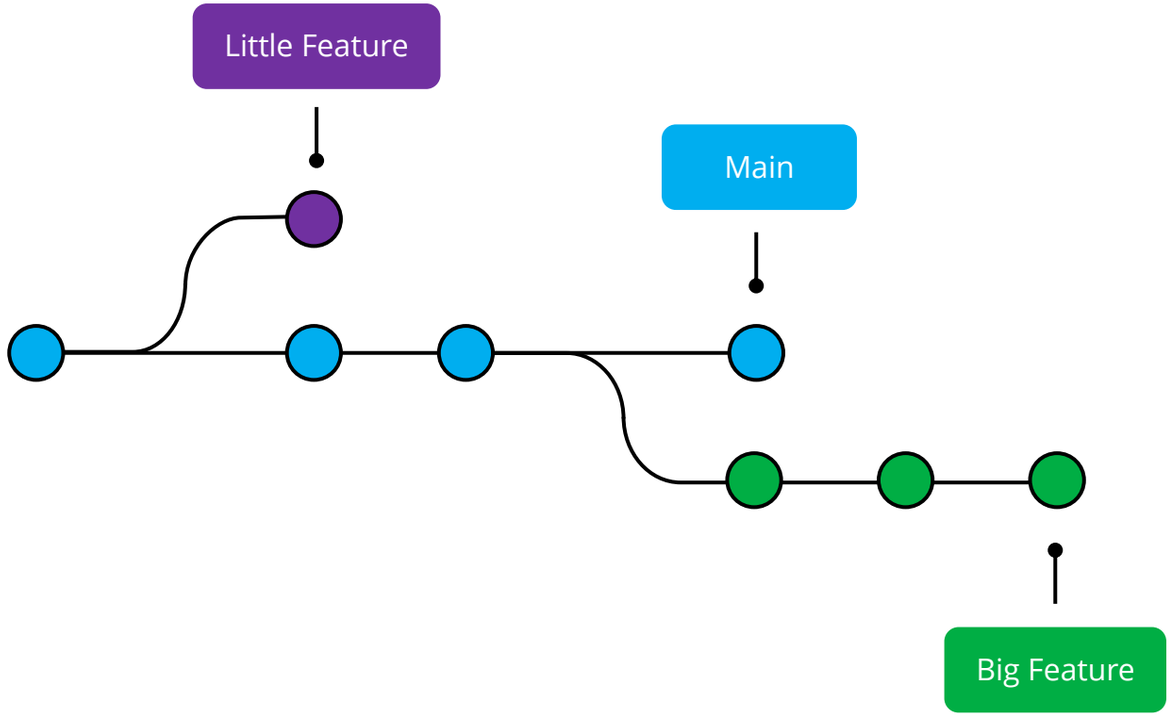
- Developers should not modify the files in the repository directly.
- They need to obtain their own copy of the codebase known as **Working Copy**.
- The developer who edits the code must “commit” their working copy and send it to the repository.
- Other team members can review/make updates.

Branching Strategies

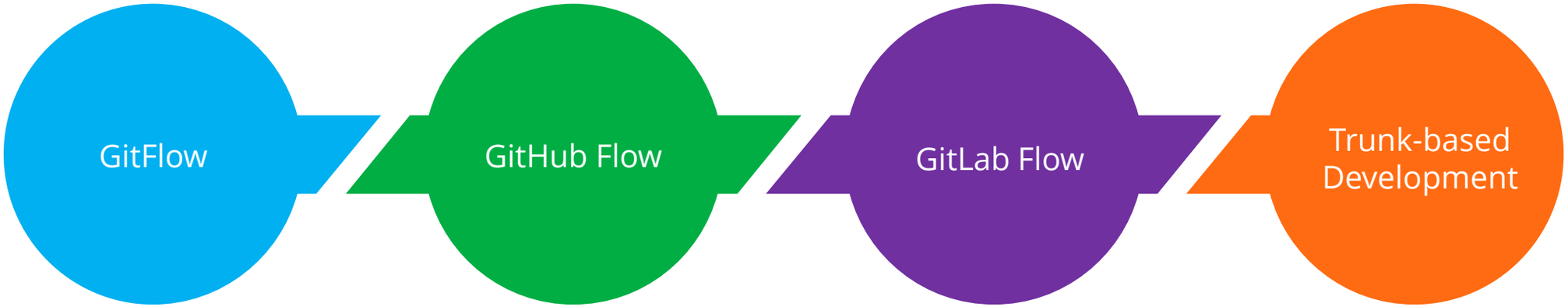
Branching is a practice of making a branch or duplicate copy of code from the mainline codebase.

Branches:

- Are used by developers to develop features giving a separate workspace for their code.
- Are used by developers to make edits to the codebase.
- Are usually merged back to a master branch upon completion of work.
- Protect the mainline from disruption or mistakes, allowing the code to function normally while undergoing updates.
- Allow other developers to work on the code concurrently without interfering with each other.



Branching Strategies



Every strategy has pros and cons and should be chosen depends on the use case.

Uses simple version of GitFlow with main “develop” branch and release/hotfix branches.

Check Your Understanding

Match the components with their corresponding Features.

*Drag and drop each component next to its feature, and then click **Submit.***

Copies of the mainline codebase	Own copy of the codebase	Enables tracking and controlling the changes to files and documents	Core of the VCS
VCS	Repositories	Working Copy	Branches

SUBMIT



Check Your Understanding

What are the different types of branching strategies?

*Select the correct answers, and then click **Submit.***

- [ ] GitFlow
- [ ] GitHub Flow
- [ ] GitLab Flow
- [ ] Git-based development

SUBMIT



KEY TAKEAWAYS

GIT is a collaborative workspace where developers can track the version of the source code.

VCS is a system that enables tracking and controlling the changes to files and documents.

Repositories is the core of the VCS that house the edited code.

Working copy is located on a developer's computer.

Branching is a practice of making a duplicate copy of code from the mainline codebase.

Branches are isolated copies of the mainline codebase that developers create to make edits.